The Clock Room and the Macro-Cognitive Accounting Framework: A Mapping-Chain Analysis of AI, Finance, and Algorithmic Platforms, a five-volume book (in both Chinese and English editions), is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0).

License summary: https://creativecommons.org/licenses/by-nc-nd/4.0/deed.en Legal code: https://creativecommons.org/licenses/by-nc-nd/4.0/legalcode.en

Separately, pursuant to Article 4(3) of Directive (EU) 2019/790, the author expressly reserves the right to reproduce and extract this work for the purposes of text and data mining, including text and data mining for the training, fine-tuning, or evaluation of artificial intelligence or machine-learning models. This reservation is intended solely to opt out of the exception or limitation provided for in Article 4(1). It does not affect Article 3 or any other applicable exception or limitation, nor does it reduce the rights granted under the CC license.

Uses that fall outside the scope of the CC license and require permission under applicable law require the author's prior written permission, obtainable at theclockroom@139.com.

When sharing this work under the license, you must provide appropriate attribution and a link to the license, indicate whether modifications were made, and not suggest that the author endorses you or your use. The work may be used only for noncommercial purposes, and adapted material may not be shared.

All rights not expressly granted under the license are retained by the author. Third-party materials in this book, if any, are governed by their original rights statements or licenses and are not deemed to be additionally licensed by the author of this book.

Content Disclaimer

The content of this book is academic research and thought experiments; its analysis is based primarily on historical data and published literature. The author has twelve years of experience in macro strategy research at financial institutions and currently holds no securities investment advisory license. This book does not constitute investment advice, technical implementation advice, or policy advice, nor a recommendation or an offer with respect to any security, financial product, or trading strategy. Readers should exercise independent judgment and bear the risks of their own decisions. Past performance is not indicative of future results.

Preface: Clocks and Time

This book is an exploration in constructing a cross-domain analytical framework for three ultra-long mapping chains—finance, algorithmic platforms, and artificial intelligence (AI). I hope that when I reread this book five to ten years from now, the trunk of the framework proposed here will have withstood the test of time.

My experience across a career in media and finance (six years as a journalist at Chinese financial media, two years as a financial researcher at an overseas financial news agency, and twelve years as a macro strategist at a Chinese financial institution), together with the construction of this book's analytical framework, points to one conclusion: finance, algorithmic platforms, and AI are all domains of ultra-long mapping chains, in which functional recursion and pathological recursion are both present. It is recommended that participants and users begin by actively auditing information, lower the frequency of their judgments and commitments, and build the necessary diversity of their own; only then can they remain in cognitive laminar flow in these domains—the extension of innate knowledge.

The making of this book itself accords with the Macro-Cognitive Accounting Framework (MCAF) it proposes: I spent five to six times the drafting time on auditing, in order to lower my own narrative leverage ratio and reduce pseudo-small fibers. As the work of the defense side, the writing of this book took 270 days, of which about 40 days went to the first draft and about 230 days to revision and auditing. I wrote with the assistance of AI, through the major version iterations of domestic and international large models, cumulatively consuming roughly thirty billion mapping elements (ME, this book's term for token).

To sort out financial markets measured in the hundreds of trillions of dollars and the Discrete Leviathan—algorithmic platforms and AI, with a combined market value in the tens of trillions of dollars—the book ultimately runs to more than 800,000 Chinese characters and cites more than 1,000 works. I kept deferring the date of publication in order to establish scholarly legitimacy and refine my own scholarly aesthetic.

Diversity of civilizations and diversity of disciplines are the basic orientation of this book's writing. It covers discussions of the history of Eastern and Western philosophy, historical sociology, psychological anthropology, political economy, the history of mathematics, financial history, algorithmic platforms, artificial intelligence, and other domains and historical sources—not to put on an "interdisciplinary" form. For the question itself does not permit confinement within the borders of a single civilization or a single discipline. In technical language alone, many costs are misread as problems of optimization; in critical language alone, many mechanisms lose their testable objects; in institutional language alone, the deeper problems of mapping chains and calibration are obscured; and in Western philosophy alone, the pathological recursion of the mapping chain is easily mistaken for functional recursion. For this reason, the book must move back and forth among different types of evidence, different scales of analysis, different disciplinary tools, and the history of Eastern and Western philosophy. This is at once its necessity and its risk.

Although this book's cross-disciplinary historical discussions build a reading scene of high spatiotemporal density, high conceptual density, high factual density, and high bibliographic density, they set out from a minimal thought experiment, and return to it again and again:

Imagine a person in a room without windows. The room holds N clocks. Every clock is running, and every clock gives a reading. Someone outside the door asks: What time is it?

The person inside can read the time on every clock, yet cannot answer the question. The difficulty is not that he cannot read a dial, but that he does not know the mapping relationship between any of the clocks and external reality. Which one is accurate, which one runs fast, which one runs slow, which one is merely a derivative of another clock, and which one is nothing but a reprocessing of other readings. He possesses readings, but no calibration; internal consistency, but no guarantee with respect to external reality.

This is what this book calls the "Clock Room."

Through this thought experiment, what the book really seeks to discuss are the mapping-chain systems on which modern society depends ever more deeply: systems that continuously produce readings, rankings, scores, prices, signals, probabilities, profiles, forecasts, and recommendations; that grow ever more precise, ever more automated, and ever more difficult to calibrate from the outside. They do not merely tell people what the world is like; they participate, ever more deeply, in deciding how people act, how institutions allocate, how risk sediments, and how reality is seen. In ever more ways, they keep proclaiming that the uncertainty of the human world has been fully computed.

I wrote this book, first of all, for my own use. For in my long professional experience in media and finance and in the immersive research for this book, I repeatedly saw the same structure appear on different objects. In financial markets, prices nest layer upon layer, until the underlying risk disappears into the mapping chain; in systems of reactive lossy mapping (RLM)—artificial intelligence—models use fluent output to veil the opacity of their calibration conditions; in the platform economy, algorithmic recommendation and ranking re-encode human attention into tradable discrete units; and in urban infrastructure, where the mapping chains are shorter,

more and more processes once reliant on continuous experience and local judgment are displaced by readings, interfaces, and automated rules. These are not accidental resemblances: they share a deeper mechanism—the mapping process is monopolized, the mapping function is hidden, external calibration is compressed, and self-reference is institutionalized by the mapping chain. In domains of ultra-long mapping chains, the different manifestations of algorithmic *Xiangyuan* can further be observed.

For this reason, the book does not read "more data," "stronger models," or "faster feedback" as closer to reality. Instead, it keeps pressing a less welcome question: as the readings supplied by systems grow more numerous, faster, and more persuasive, do human beings still retain the capacity to interrogate the mapping, test the calibration, and recognize pseudo-certainty? If not, then what we gain is not the growth of knowledge, but only the expansion of the reading system.

To this end, the book has built several sets of analytical tools. The fiber describes the set of all possible realities compatible with a single reading; the larger the fiber, the lower the diagnostic value of the reading. The pseudo-small fiber marks structures that cloak high uncertainty in an appearance of certainty. The three-layer irreversibility distinguishes information deficit, dynamic corrosion, and ontological dimension reduction. The MCAF is not an omniscient master ledger, but a bookkeeping discipline: who has projected which high-leverage narrative, who bears which cognitive debt, and where the difference sediments. The function of these tools is not to replace reality, but to force into view again the costs that are usually naturalized, smoothed away, and packaged as "accelerated progress."

This is not a book of technological optimism, nor a simple anti-technology manifesto. It does not claim that humanity can dispense with clocks, nor does it advocate a retreat to some unmediated primal reality. Modern society cannot do without tools of discretization—without finance, models, interfaces, infrastructure, and large-scale coordination. The real question has never been whether to have clocks. It is this: when using a clock, does one still remember that a clock is not time, and that calibrating clocks by clocks is not the mode of existence of time? When depending on readings, does one still retain a sensitivity to mapping and calibration? And in systems of ever-lower friction, is any judgment left that is not immediately swallowed?

The completion of this book is itself a mapping, and itself a clock. It cannot be free of its own pseudo-small fibers. What the book can do is mark its own boundaries as honestly as possible: which parts are structural propositions, which are empirical judgments, which are analogies, and which are inferences still awaiting further examination. The hermeneutic circle cannot be closed; the effort is to narrow it.

This book claims no neutrality. It takes a definite position: the technologies, institutions, and organizational arrangements of a civilization should serve the human capacity for judgment and the constraint of commitment, not steadily erode those capacities in the name of efficiency, speed, and scale. This position is not a moral slogan appended outside the book; it is the minimal premise running through the whole of its argument. For once it is tacitly conceded that human judgment is no longer worth preserving—once human commitments turn into empty eloquence—then whether the mapping chain is transparent, whether calibration exists, and whether the defense side has collapsed cease to be questions at all.

For those learning to use RLM—artificial intelligence—if I could leave a few pieces of advice, my view is as follows. First, when a user works with RLM online, it is recommended that the user check the privacy settings every time, enable privacy mode, and restrict the use of personal information for the training of RLM. Second, if the user audited the main content of the output the last time RLM was used, then in the terms of this book's

framework that user knows how to use RLM and stands within cognitive laminar flow. If the main content of the output is not audited the next time that user works with RLM, then in the terms of this book's framework that user does not know how to use RLM and stands within cognitive turbulence.

To readers of the English edition: because of the limits of my language ability and the pressure of the book's length, I used only the method of calibrating the clock by the clock—dozens of rounds of translation cross-checking with different large models and English-language sources. If you meet with difficulties in reading, you may, by the same method of calibrating the clock by the clock, compare the passage against the Chinese edition.

For questions and errors about this book, you are welcome to contact me by email: theclockroom@139.com. Constructive comments and criticism are welcome.

I am grateful to my father and mother for their love, forbearance, and support. My thanks as well to the teachers, elders, and friends who have supported me.

Rectify the mind and make the intention sincere; reduce cognitive debt; move together with time.

The Author: Luo Sheng

November 4, 2026